CREDIT CARD FRAUD DETECTION

Using Machine Learning & Anomaly Detection

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DECLARATION

We declare that this written submission represents our ideas in our own words and where other's ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all the principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission.

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Date: 23 April 2026

CERTIFICATE

This is to certify that the project report titled "Credit Card Fraud Detection" has been submitted by the following students as part of their coursework for the degree of Bachelor of Technology (B.Tech) at K.R. Mangalam University.

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This work has been carried out under our supervision and has not been submitted anywhere else for any other degree or examination.

Name of Supervisor: Mr. Kunal Rai

Date: 23 April 2026

Place: K.R. Mangalam University

ACKNOWLEDGEMENT

"Enthusiasm is the foundation of all progress — with it there is accomplishment, and without it there are only excuses."

We would like to take a moment to sincerely thank everyone who helped us throughout this project. When we first started, it felt quite overwhelming — especially since we were working with tools and techniques that were new to us. But with the right guidance and support, we were able to complete it successfully.

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EXECUTIVE SUMMARY / ABSTRACT

Credit card fraud has become a major concern in today's digital economy, where millions of financial transactions are performed every day. Due to the rapid increase in online payments, detecting fraudulent transactions in real time has become essential for both financial institutions and customers.

This project presents an effective system for detecting fraudulent credit card transactions using Machine Learning techniques. The dataset used is highly imbalanced, with a very small percentage of fraudulent cases, making the problem more challenging. To address this, various data preprocessing and class imbalance handling techniques such as undersampling, oversampling, and SMOTE were applied.

Multiple machine learning models including Logistic Regression, Decision Tree, Random Forest, and XGBoost were implemented and evaluated using performance metrics such as ROC-AUC, Precision, Recall, and F1-score. Among these, XGBoost achieved the best performance, providing high accuracy and strong fraud detection capability.

The results demonstrate that machine learning models can effectively identify fraudulent patterns in transaction data and help minimize financial losses. This system can be further extended to real-time fraud detection and integrated into banking systems for enhanced security.

Key Words: Machine Learning, Fraud Detection, Imbalanced Data, XGBoost, SMOTE, Financial Security.

1. INTRODUCTION

1.1 Background

In today's digital economy, the use of credit cards and online transactions has increased significantly. While this growth has made payments faster and more convenient, it has also led to a rise in fraudulent activities. Credit card fraud, which involves unauthorized or illegal transactions, has become a major challenge for financial institutions and customers worldwide.

With millions of transactions occurring every day, it is difficult to manually identify fraudulent activities. Traditional rule-based systems are no longer sufficient to detect complex and evolving fraud patterns. This is where Deep Learning plays a crucial role.

This project focuses on building an intelligent system that can automatically detect fraudulent transactions using Neural Networks (NN) and Convolutional Neural Networks (CNN). By analyzing transaction patterns and behavior, the model can distinguish between legitimate and fraudulent activities.

1.2 Problem Statement

With the rapid growth of digital payments, credit card fraud has become increasingly sophisticated and difficult to detect. Fraudulent transactions often represent a very small portion of the total data, making the dataset highly imbalanced. This imbalance creates challenges for accurate prediction.

Manual monitoring of transactions is inefficient and cannot scale with the growing volume of data. Therefore, there is a need for an automated system that can quickly and accurately identify fraudulent transactions using advanced techniques like Neural Networks and CNN.

1.3 Objectives

Primary Objective

To develop a deep learning model using Neural Networks (NN) and Convolutional Neural Networks (CNN) for detecting fraudulent credit card transactions.

Specific Objectives

- Analyze and understand the dataset
- Perform data preprocessing and feature scaling
- Handle class imbalance using SMOTE
- Build a Neural Network (ANN) model
- Build a Convolutional Neural Network (CNN) model
- Compare the performance of both models

2. DATA DESCRIPTION & TECHNICAL STACK

2.1 Data Dictionary

The dataset used in this project is sourced from publicly available financial transaction data from Kaggle. It contains anonymized credit card transaction records. Due to privacy concerns, the features are transformed and encoded.

Feature	Description	Data Type
amt	Transaction amount	Numerical
Gender	Gender of cardholder (encoded)	Numerical
Lat, long	Location coordinates	Numerical
City_pop	Population of city	Numerical

2.2 Technical Stack

The following tools and technologies were used throughout this project:

Purpose	Tools & Libraries
Programming Language	Python
Data Handling	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Deep Learning	TensorFlow, Keras
Model Types	Neural Network (ANN), CNN

2.3 Data Source

The dataset was obtained from Kaggle's Credit Card Fraud Detection repository. It contains a large number of transactions with a highly imbalanced distribution, where fraudulent transactions represent a very small percentage of the total data. This dataset is widely used for building and evaluating fraud detection models.

3. EXPLORATORY DATA ANALYSIS (EDA)

3.1 Data Cleaning

Before building the model, the dataset was carefully preprocessed to ensure data quality and consistency.

The following steps were performed:
• Checked for missing values in all columns. The dataset did not contain any missing values.
• Removed unnecessary features such as IDs, names, and irrelevant columns that do not contribute significantly to fraud detection.
• Converted categorical variables into numerical form using encoding techniques.
• Scaled the numerical features using StandardScaler to normalize the data.
• Handled class imbalance using SMOTE to improve model performance.
• Ensured that all features were in a suitable format for deep learning models.

3.2 Descriptive Statistics

The dataset is highly imbalanced, with fraudulent transactions representing only a small fraction.

Key observations include:
• Fraud transactions account for a very small percentage of the dataset.
• Most transactions are non-fraudulent, creating a strong class imbalance.
• Transaction amounts vary across different ranges.
• Features include transaction amount, location, category, and user-related information.

3.3 Correlation Analysis

To understand relationships between features and the target variable:
• Correlation analysis was performed to identify patterns between features and fraud labels.
• Certain features showed relationships with fraudulent transactions.
• No strong multicollinearity issues were observed that could significantly affect model performance.

3.4 Data Visualization

Visualization techniques were used to better understand the dataset:
• Class Distribution Plot: Showed imbalance between fraud and non-fraud transactions.
• Transaction Amount Distribution: Highlighted spending patterns.
• Feature Distribution Plots: Displayed variation in different variables.
• Correlation Heatmap: Helped identify relationships between features.

These visual insights helped in selecting appropriate preprocessing techniques and models for fraud detection.

MODEL DEVELOPMENT & ANALYSIS

4.1 Feature Selection

The dataset consists of transaction-related features such as amount, location, category, and user information. Since the data was already preprocessed and encoded, explicit feature selection was not required.

However, the following preprocessing steps were applied:
• Removed unnecessary features that did not significantly contribute to fraud detection.
• Scaled the numerical features using StandardScaler.
• Handled class imbalance using SMOTE.
• Used all processed features as input for model training.

4.2 Model Selection

Since this is a binary classification problem with highly imbalanced data, deep learning models were implemented and compared:

• Neural Network (ANN)
• Convolutional Neural Network (CNN)

The dataset was split into 80% training and 20% testing. CNN was selected as the final model due to its slightly better performance and ability to capture complex patterns.

4.3 Model Performance

The models were evaluated using appropriate metrics for imbalanced data:

Metric	Result
ROC-AUC	Measures model performance across all thresholds
Precision	Measures accuracy of fraud predictions
Recall	Measures ability to detect fraud cases
F1 Score	Balance between precision and recall

The model demonstrated strong capability in identifying fraud patterns despite extreme class imbalance.

Model Code

```
# Import Libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Load Dataset
df = pd.read_csv("creditcard.csv")

# Split Features & Target
X = df.drop("Class", axis=1)
y = df["Class"]

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Model (ANN)
model = Sequential([
    Dense(16, activation='relu', input_shape=(X_train.shape[1],)),
    Dense(8, activation='relu'),
    Dense(1, activation='sigmoid')
])

# Compile
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])

# Train
model.fit(X_train, y_train, epochs=10, batch_size=512, validation_split=0.2)

# Evaluate
loss, acc = model.evaluate(X_test, y_test)
print("Accuracy:", acc)
```

5. RESULTS & DISCUSSION

5.1 Key Insights

The project successfully developed a Credit Card Fraud Detection System capable of identifying fraudulent transactions with high accuracy. The following are the key insights derived from the analysis and model results:

5.1.1 Imbalanced Data Challenge: The dataset was highly imbalanced, with fraudulent transactions accounting for a very small portion of the total data. Special techniques such as SMOTE were used to handle this imbalance.

5.1.2 Effectiveness of Preprocessing: Scaling significantly improved model performance by normalizing feature distributions.

5.1.3 Model Performance Comparison: Deep learning models were evaluated, including Neural Network (ANN) and Convolutional Neural Network (CNN).

5.1.4 XGBoost Superiority: CNN slightly outperformed ANN in terms of loss and overall performance, showing better capability in capturing complex patterns.

5.1.5 Importance of Recall: High recall was critical in fraud detection to minimize false negatives (i.e., undetected fraud cases).

5.2 Visual Dashboards

Throughout the project, various visualizations were used to better understand the dataset and model performance:

5.2.1 Class Distribution Plot: Highlighted the extreme imbalance between fraudulent and non-fraudulent transactions.

5.2.2 Transaction Amount Distribution: Showed differences in transaction amounts between fraud and normal cases.

5.2.3 Correlation Heatmap: Helped identify relationships between different features.

5.2.4 Confusion Matrix: Displayed model performance in terms of true positives, false positives, true negatives, and false negatives.

6. CONCLUSION & RECOMMENDATIONS

6.1 Summary of Findings

The project demonstrated that:

6.1.1 Fraud detection is feasible using deep learning techniques on transactional data.
6.1.2 Handling class imbalance (using SMOTE, etc.) significantly improves model performance.
6.1.3 Feature scaling and preprocessing enhance the effectiveness of deep learning models.
6.1.4 Models like Neural Networks and CNN perform effectively in detecting fraud patterns in tabular data.

6.2 Strategic Recommendations

Based on the project outcomes, the following recommendations are proposed:

6.2.1 Real-Time Monitoring: Integrating the model into real-time transaction systems can help detect fraud instantly.
6.2.2 Model Retraining: The model should be periodically retrained with new transaction data to adapt to evolving fraud patterns.
6.2.3 Multi-Layer Security: The model should be used alongside other security mechanisms (e.g., OTP verification, behavioral analysis).
6.2.4 Risk Scoring Systems: Implementing a risk score for each transaction can help prioritize suspicious activities.

6.3 Future Scope

Several improvements can further enhance this system:

6.3.1 Advanced Models: Implementing more advanced deep learning models for anomaly detection.
6.3.2 Deployment: Developing a web application using Flask or FastAPI for real-time usage.
6.3.3 Real-Time Streaming: Integrating with streaming platforms for continuous fraud detection.
6.3.4 Explainability: Using techniques to understand model decisions.
6.3.5 Hybrid Systems: Combining deep learning models with rule-based systems for better accuracy.

7. REFERENCES & APPENDIX

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